UML Use Case: Sudoku System

1. Brief Description

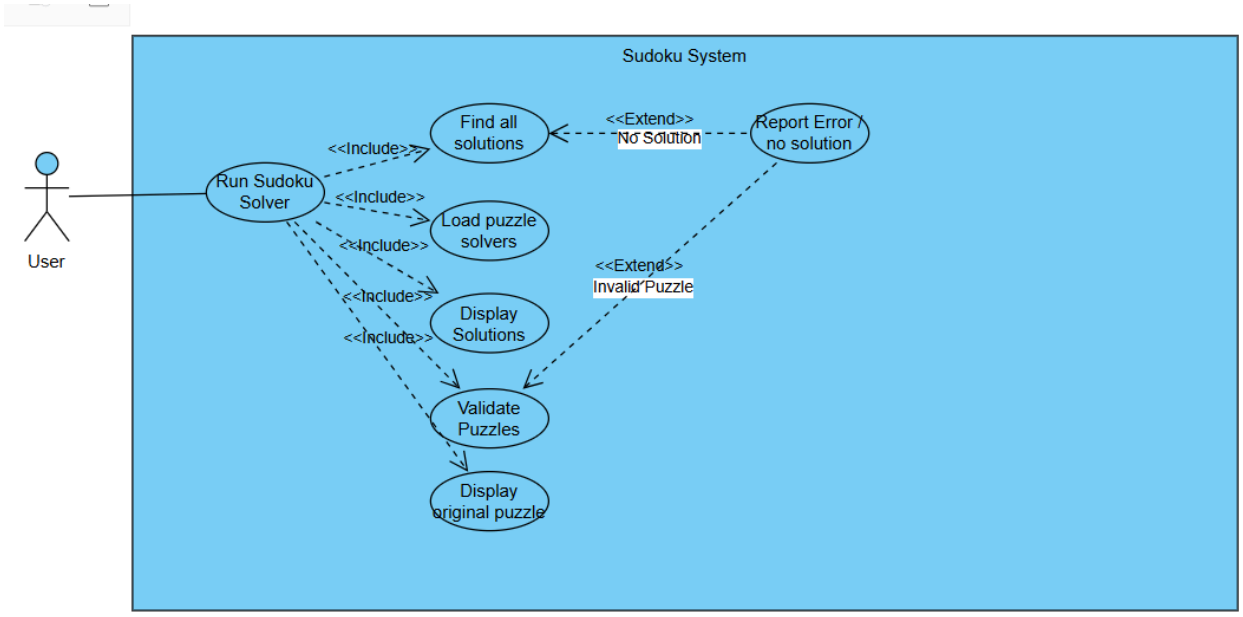
The user runs a Sudoku-solving program that reads five puzzle files. The program checks each puzzle, displays the original board, and uses recursive backtracking to find and print all possible solutions. If a puzzle cannot be solved, the program displays "No solution found."

2. Step by Step Flow

- The user starts the Sudoku Solver program.
- The system attempts to load a Sudoku puzzle file.
- The system checks that the file contains 81 valid cells.
- The original Sudoku puzzle is displayed.
- The system checks whether the starting numbers follow Sudoku rules.
- The program uses recursive backtracking to find every possible solution.
- Each solution is displayed to the user.
- If no solution exists, the program displays "No solution found."
- The program repeats this process for all five puzzle files.

3. Detailed Use Case Description

Use Case Name: Run Sudoku Solver
Primary Actor: User
Preconditions: The program is compiled and the five Sudoku puzzle files are available in the correct directory.
Main Flow: The user starts the program, which automatically loads each puzzle file. The system checks the file format and starting numbers before solving the puzzle. Using recursive backtracking, the program tries valid numbers in empty cells until it finds every possible solution. It then displays the completed Sudoku boards.
Alternative Flow: If a file cannot be opened or contains invalid characters or an incorrect number of cells, the program displays an error message and continues to the next file. If the puzzle has conflicting numbers or cannot be solved, the program displays "No solution found."
Postconditions: All five puzzle files have been processed, and the available solutions or appropriate error messages have been displayed.


Sudoku System
Find all solutions
<<Extend>>
No Solution
Report Error / no solution
<<Include>>
Run Sudoku Solver
<<Include>>
Load puzzle solvers
<<Include>>
<<Include>>
Display Solutions
<<Include>>
<<Extend>>
Invalid Puzzle
Validate Puzzles
Display original puzzle
User